Sector-ETF Momentum with a Volatility-State Gate: an ML4T Midterm Proposal

Setup v1 | 23 sector/industry ETFs + QQQ, USO, GLD, 2019-2026 (Yahoo) | Development 2020-01..2025-12, sealed holdout 2026-01..2026-10 (fresh data, touched once) | Repository: ml4t-midterm-sector23 (git 7202343 at setup)

Abstract. We test whether sector and industry ETF trends, traded only outside high-volatility states, predict next-month returns, and whether Ridge or LightGBM improves on a momentum rule. Under a pre-registered walk-forward protocol (8 half-year folds, 2022-2025) no model beat the rule, so the rule was deployed (validation IC +0.010 [-0.057, +0.078], net excess Sharpe 0.36); momentum had no IC in calm states and the gate made the rule worse, falsifying our mechanism on validation. On the fresh 2026 holdout the deployed strategy earned an excess Sharpe of 0.07 [-2.01, 1.45] against 1.88 for the equal-weight universe; the pre-registered confirmatory test failed (-1.816, CI -3.602..+0.498). Outcome pattern: **never resolved**.

1. Hypothesis and Sources of Edge

Mechanism (two sentences). Industry and sector returns persist over one to six months because information diffuses slowly across industries and investors rebalance sector allocations gradually. The persistence crashes when market volatility spikes and leadership reverses abruptly, so the rotation signal should be traded only in calm-to-normal volatility states.

Source of edge (Table 6.1). Flow predictability (gradual sector re-allocation by benchmarked and thematic investors) and risk compensation (holding leading risk sectors earns a premium that is lost in volatility shocks). **Counterparty:** investors who rebalance on calendar schedules or chase sector leadership late.

Falsification criteria (fixed in setup.yaml before data analysis). (1) The momentum rule's validation IC is not positive in the low-volatility state. (2) The gated rule does not beat the ungated rule on validation net Sharpe. (3) The pre-registered confirmatory test fails on the holdout: deployed holdout net excess Sharpe minus the equal-weight benchmark, 21-day block bootstrap, pass only if the 95% CI lower bound is above zero.

Status. (1) **triggered** (low-vol IC -0.013, high-vol +0.089 - the opposite of the mechanism); (2) **triggered** (gated rule 0.36 vs ungated 0.41); (3) **triggered** (Section 7). The hypothesis is not supported; the report documents where it failed.

What could break this? Six years cover few regimes (post-COVID recovery, 2022 inflation shock, 2023-24 AI rally, 2025 tariff shock). With 23 overlapping ETFs the cross-section is thin, and the 80th-percentile gate threshold was chosen a priori, not estimated.

2. Trading Setup v1 and Evaluation Protocol

Setup field	Specification (setup.yaml, committed before any backtest)
Universe and eligibility	Fixed list of 23 ETFs chosen by the team: IYR, XLB, XLC, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY, QQQ, IBB, ITA, ITB, KRE, OIH, SOXX, XBI, XME, XRT, USO, GLD. Point-in-time eligibility: trailing 126-session median dollar volume >= $10M and 126 sessions of history at t (all 23 qualify at every decision). No de-duplication: overlaps such as QQQ/XLK (corr 0.98) are kept and disclosed. BIL is cash; SPY, TLT, AGG are used for states and benchmarks only.
Decision schedule	Snapshot at the month-end close; execute at the next session's open; hold to the next rebalance (about 21 sessions).
Admissible information	Prices and volume to the close of t; no fundamentals; longest lookback 147 sessions.
Score-to-trade mapping	Rank, top-5, equal weight (inverse-volatility as the comparison allocator), long-only. No-trade state: SPY 21-day vol percentile vs its trailing 126 sessions >= 0.80 -> 100% BIL.
Constraints and risk	Leverage <= 1, no shorting, max weight 20%, checked inside the simulation. Overlay: 10% annual volatility target from 63-session returns, remainder in BIL.
Cost-model class	$0.0035/share commission + tiered half-spread ($0.005 SPY/QQQ, $0.01 sector XL*, $0.02 other); sweep 0-8x; stress 3x spreads.
Version	setup_version: v1

Protocol commitment	Value
Training window	expanding from 2020-01-02
Step / retraining cadence	6 months / 6 months
Label horizon	21 sessions, open[t+22] / open[t+1] - 1
Longest feature lookback	147 sessions
Validation folds	2022-01-01..2022-06-30; 2022-07-01..2022-12-31; 2023-01-01..2023-06-30; 2023-07-01..2023-12-31; 2024-01-01..2024-06-30; 2024-07-01..2024-12-31; 2025-01-01..2025-06-30; 2025-07-01..2025-12-31
Label buffer (purge) / feature buffer (embargo)	21 / 0 trading sessions
Hyperparameter tuning	fixed_holdout (walk-forward validation folds only; never the holdout)
Holdout (sealed)	2026-01-01..2026-10-30
Yardstick	mean daily Spearman IC with Newey-West 95% CI (lag 21)
Confirmatory test	holdout net Sharpe of deployed strategy minus equal-weight benchmark Sharpe (paired, 21-day block bootstrap); pass if 95% CI lower bound > 0

Why the holdout is 2026. Earlier versions of this project had already looked at 2024 and 2025 outcomes, so those years cannot be an honest final test; they are development data here. The holdout is 2026-01-02..2026-10-30, downloaded from Yahoo with the textbook's own source; no one on the team had looked at it when setup.yaml was committed, and the loader refuses to read it before the freeze. **Engine.** A custom monthly event loop (src/lib.py: simulate): the gate, the volatility overlay and the constraints use information at the close of t and change the next action. Fills are at the next open; a fill on a missing bar raises an error. **Search accounting.** The research ledger (Appendix A) was opened by 01_setup before any backtest; run_log.csv records git commit, data snapshot, configuration, window, metric and decision for every run. This proposal tried 60 strategy variants; including the project's three earlier versions the inclusive count is 906.

What could break this? The 23-ETF list was chosen by the team after earlier versions had shown 2020-2025 results, a hindsight risk in the universe that the protocol cannot remove. The holdout has only ten monthly decisions.

3. Data: Sources, Point-in-Time Pipeline, Quality Report

Route A (textbook ETF universe, extended). All 27 symbols are in the textbook's ETF dataset; we re-downloaded them from Yahoo (yfinance) for 2019-01-02..2026-10-30 so that a never-seen 2026 holdout exists. **Timestamp semantics:** bar date = NYSE session; close known at 16:00 New York, open at 09:30. **Corporate actions:** split- and dividend-adjusted OHLCV for returns and fills; split-adjusted, dividend-unadjusted close for dollar volume and per-share costs. **Identifiers:** ticker = key; no ticker changes in the period. **Storage:** data/prices_adjusted.parquet and data/prices_raw.parquet, rebuilt by data/build_dataset.py (download time in data/download_meta.json).

Quality check (development 2019-2025)	Result
Symbols x sessions; duplicate symbol-dates	27 x 1760; 0
Missing values (open/high/low/close/volume)	0 / 0 / 0 / 0 / 0 (no forward-fill; the simulator refuses to fill on a missing bar)
OHLC invariant violations	high_below_low 0, close_outside_high_low 0, open_outside_high_low 0, non_positive_price 0, zero_volume 0
Daily moves > 15%	17, all in March-November 2020 except SOXX +18.6% on 2025-04-09 (tariff pause); verified as real and kept
Source consistency	Daily adjusted returns vs the textbook's Yahoo bars, 40,689 overlapping rows: correlation 1.00000000, share differing by > 10 bp 0.0%
Corporate-action check	USO 1:8 reverse split 2020-04-29: largest unadjusted daily change 745%, adjusted 25% (Figure D2)
Survivorship	All 23 names trade through 2026: a hand-picked survivor list (bias disclosed)
Coverage	23-23 of 23 eligible at every decision; highest pairwise return correlations: QQQ/XLK 0.976, OIH/XLE 0.914, SOXX/XLK 0.909, IBB/XBI 0.907

PIT level	Check	Result
Values	Every feature uses bars dated <= t (close of t); label starts at the t+1 open	PASS
Values	Non-price data	N/A (prices and volume only)
Universe	Eligibility (ADV, history) computed from the trailing 126 sessions at each t	PASS
Universe	Delisted / removed ETFs included	FAIL (disclosed): fixed list of 23 survivors chosen in 2026 by the team; no delisted sector ETFs
Identifiers	Ticker changes / reuse handled with effective dates	PASS (no ticker changes in 2019-2025 for these symbols; symbols are stable keys)

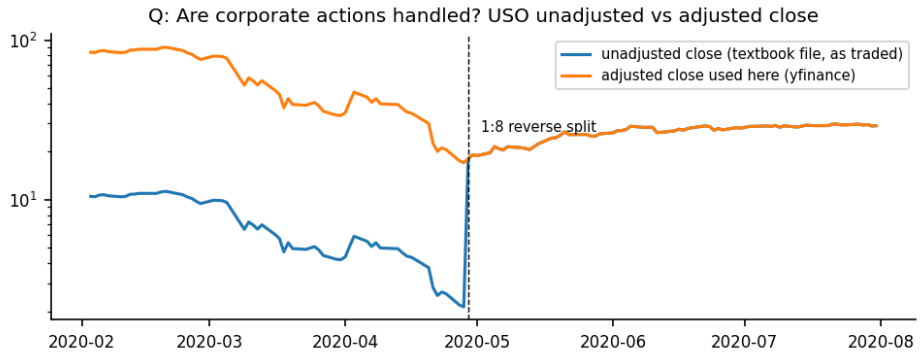


Figure D2. Q: Are corporate actions handled? The unadjusted series shows the 1:8 reverse split as an 8x jump; the adjusted series used here is continuous.

What could break this? Yahoo revises adjusted history, so a later download can differ slightly (download time recorded). Because the dividend-unadjusted close is split-adjusted, USO's per-share cost before its 2020 split is understated; no validation trade falls in that period.

4. Labels, Features, and Triage

Label. Fixed-horizon continuous return fwd_oo_21 = open[t+22]/open[t+1] - 1, anchored to the next-open fill (a close-to-close label would credit the overnight gap the strategy cannot earn). Models learn its cross-sectional percentile. **Overlap:** 34,178 nominal labelled rows over 1,486 sessions, but with a 21-session horizon only about 70 non-overlapping dates exist, each with just 23 names; every IC interval uses Newey-West with lag 21.

Feature	Role	Driver hypothesis	Lookback	Representation	Known failure mode
mom_126_vs	Signal	Persistence / under-reaction (flows)	126d, monthly	126d return / 63d vol, cross-sectional rank	Momentum crashes at volatility spikes and trend reversals
mom_126_skip21	Signal	Persistence without 1-month reversal	126d skip 21d	Return, rank	Same as momentum; crowding
mom_63	Signal	Medium-term persistence	63d	Return, rank	Choppy markets
ret_21	Signal	Short-term reversal (liquidity provision)	21d	Return, rank	Trending markets
ret_5	Signal	Weekly reversal	5d	Return, rank	Noise at a monthly hold
dist_63d_high	Signal	Anchoring to recent high	63d	Distance to max, rank	Breakouts in crises
sma50_ratio	Signal	Trend-following flows	50d	Price / SMA, rank	Whipsaw
vol_63	Signal	Risk compensation vs low-vol anomaly	63d	Annualised vol, rank	Sign depends on regime
idio_vol_63	Signal	Idiosyncratic risk premium	63d	Residual vol vs SPY, rank	Unstable beta estimate
beta_spy_63	Signal	Equity-risk compensation / betting-against-beta	63d	Beta to SPY, rank	Flips sign in sell-offs
max_dd_63	Signal	Loss aversion / distress	126d (63 within 63)	Worst drawdown, rank	Rebounds after crashes
volume_chg_21_63	Signal	Attention and flow pressure	63d	Log volume ratio, rank	Event-driven volume spikes
range_21	Signal	Risk compensation (trading range)	21d	Mean (H-L)/C, rank	Single shock dominates
rsi_14	Signal	Overbought/oversold reversal	14d (EWM)	Oscillator, rank	Strong trends stay overbought
dollar_vol_126	Signal	Liquidity premium	126d	Log median $ volume, rank	Size proxy, slow
spy_vol_pct	State	Risk environment: momentum works in calm states	21d vol ranked over 126d	Percentile vs own history	Single shock dominates
spy_trend_126	State	Market trend regime	126d	Return	Late at turning points
xs_dispersion_21	State	Opportunity set for rotation	21d	Cross-sectional std	Driven by one outlier ETF
spy_tlt_corr_63	State	Hedge regime (stock-bond correlation)	63d	Level	Non-synchronous trading

Searched set and de-duplication. 15 signals + 4 states. Correlation clustering (|Spearman| >= 0.80, pre-declared priority = table order) merged: mom_126_skip21 -> mom_126_vs, sma50_ratio -> ret_21, max_dd_63 -> vol_63, range_21 -> vol_63, rsi_14 -> ret_21.

Feature	IC [NW CI]	ICIR	Same sign	IC h5/h21/h63	Placebo	Mono t.	Q5-Q1 bp/mo	BHp	Gates G1/G2/G3/G4/G5	Decision
mom_126_vs	+0.009 [-0.051,+0.069]	0.10	50%	+0.00/+0.01/+0.01	+0.003	+0.6	13	0.85	pass/fail/non-/pass/fail	STOP
mom_63	-0.023 [-0.086,+0.039]	-0.15	58%	-0.01/-0.04/-0.05	-0.044	+0.7	-28	0.85	slow/fail/non-/pass/fail	STOP
ret_21	+0.023 [-0.032,+0.078]	0.23	58%	+0.00/+0.01/-0.04	-0.042	+0.9	119	0.85	slow/fail/non-/pass/fail	STOP
ret_5	+0.008 [-0.027,+0.042]	0.15	50%	-0.01/+0.00/-0.03	-0.017	+0.7	65	0.85	slow/fail/non-/pass/fail	STOP
dist_63d_high	-0.005 [-0.062,+0.051]	-0.06	58%	-0.00/-0.01/-0.06	-0.048	+0.1	-5	0.85	slow/fail/non-/fail/fail	STOP
vol_63	+0.026 [-0.045,+0.096]	0.29	67%	+0.01/+0.03/+0.07	+0.039	+0.7	80	0.85	slow/weak/non-/pass/fail	REVISE
idio_vol_63	-0.014 [-0.076,+0.047]	-0.12	50%	-0.01/-0.02/-0.04	-0.004	+0.3	-11	0.85	pass/fail/non-/pass/fail	STOP
beta_spy_63	+0.058 [-0.017,+0.134]	0.47	67%	+0.03/+0.06/+0.13	+0.059	+0.9	126	0.85	slow/weak/non-/pass/fail	REVISE
volume_chg_21_63	-0.009 [-0.054,+0.035]	-0.07	58%	-0.01/-0.02/-0.01	-0.008	-0.4	23	0.85	slow/fail/non-/pass/fail	STOP
dollar_vol_126	-0.010 [-0.053,+0.034]	-0.14	58%	-0.01/-0.01/-0.01	-0.014	-0.7	30	0.85	slow/fail/non-/pass/fail	STOP

Gates: G1 correctness (observable at t by construction; timing placebo = IC of the feature lagged 21 sessions; 'slow' flags a slow state proxy whose IC survives the shift); G2 IC with NW CI, ICIR over twelve half-year blocks, share of blocks with the pooled sign, decay over 5/21/63-day horizons; G3 quintile monotonicity (|rho| >= 0.9); G4 sign-aligned Q5-Q1 spread must exceed 2x the median round trip (4.7 bp); G5 Benjamini-Hochberg at 5% over

the searched set. Rule: PROCEED if G2, G3, G4, G5 pass; STOP if G2 fails with |t| < 1 or merged; otherwise REVISE. Result: 0 PROCEED, 2 REVISE, 13 STOP. No feature survives BH; the two REVISE signals (vol_63, beta_spy_63) plus the four states enter the models. The momentum rule is kept as the pre-registered baseline regardless of its triage outcome.

Top-3 feature	IC	Timing placebo	Shared driver controlled	IC after control	IC low-vol tercile	IC high-vol tercile	Event time
beta_spy_63	+0.058	+0.059	vol_63	+0.063	+0.070	+0.087	N/A
ret_21	+0.023	-0.042	beta_spy_63	+0.004	+0.005	+0.057	N/A
mom_63	-0.023	-0.044	beta_spy_63	-0.017	+0.006	-0.036	N/A

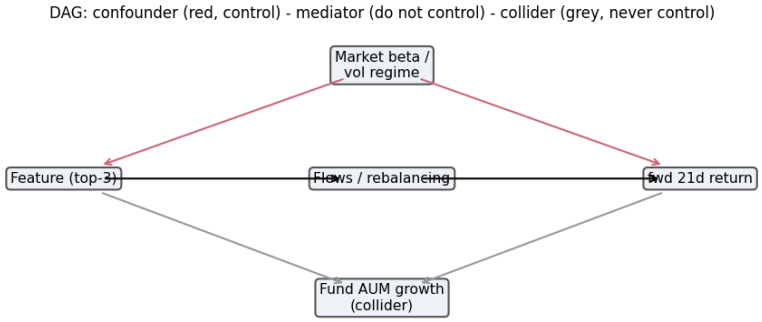


Figure D3. DAG used for the top features: market beta / volatility regime is a confounder (controlled in the shared-driver check); flows are the mediator (not controlled); fund AUM growth is a collider (never controlled).

Reading. The strongest feature is market beta (IC +0.058), a risk-compensation variable, not trend: its IC survives the 21-day timing placebo (a slow state proxy), survives a volatility control (+0.063) and is larger in high-volatility terciles. The 21-day return's IC disappears once beta is controlled (+0.004). Momentum's IC is negative in calm states and positive only in volatile ones (low-vol -0.013, high-vol +0.089) - the reverse of what the gate assumes.

What could break this? Triage used the whole development period, including the validation folds, so feature screening is not out-of-sample (only STOP removes features, which limits the leak). Beta's positive IC is a risk-on effect of 2020-2025 and can reverse in a bear market.

5. Models, Validation, Interpretation, Uncertainty

Ladder (same features, folds, purge). (1) Non-ML rule: vol-scaled 126-day momentum. (2) Ridge, alpha 1e-2..1e5 (eight values); signals enter as per-date percentile ranks, states are winsorised (1/99) and standardised with training-fold statistics only. (3) LightGBM, L1 loss, depth <= 4, 7/15 leaves x 300/1000 minimum leaf size, early stopping (50 rounds) on a purged inner split (last 126 training sessions). (4) Ensemble: mean of the three rungs' per-date z-scores. Selection by the yardstick (mean daily Spearman IC, NW 95% CI); hyperparameters never touched the holdout.

Fold	Validation	Training from	Last training label ends	Training rows	Scored rows
0	2022-01-01..2022-06-30	2020-01-02	2021-12-31	11109	2875
1	2022-07-01..2022-12-31	2020-01-02	2022-06-30	13961	2944
2	2023-01-01..2023-06-30	2020-01-02	2022-12-30	16882	2875
3	2023-07-01..2023-12-31	2020-01-02	2023-06-30	19734	2921
4	2024-01-01..2024-06-30	2020-01-02	2023-12-29	22632	2875
5	2024-07-01..2024-12-31	2020-01-02	2024-06-28	25484	2967
6	2025-01-01..2025-06-30	2020-01-02	2024-12-31	28428	2829
7	2025-07-01..2025-12-31	2020-01-02	2025-06-30	31234	2967

Model	Validation IC [NW 95% CI]	t	ICIR	Folds > 0
rule_mom126vs (chosen)	+0.010 [-0.057, +0.078]	0.30	0.16	4/8
ridge_a0.01	-0.019 [-0.107, +0.069]	-0.42	-0.14	5/8
ridge_a10	-0.019 [-0.107, +0.069]	-0.43	-0.15	5/8
ridge_a10000 (chosen)	-0.011 [-0.105, +0.084]	-0.22	-0.06	4/8
ridge_a10000 (chosen)	-0.011 [-0.105, +0.084]	-0.22	-0.06	4/8
lgb_l7_m300	-0.069 [-0.128, -0.010]	-2.30	-0.60	3/8
lgb_l7_m1000 (chosen)	-0.039 [-0.096, +0.019]	-1.30	-0.36	3/8
lgb_l15_m300	-0.083 [-0.144, -0.022]	-2.67	-0.85	3/8
lgb_l15_m1000	-0.048 [-0.107, +0.012]	-1.57	-0.39	4/8
ensemble (chosen)	-0.013 [-0.081, +0.056]	-0.36	-0.16	4/8

Paired comparison (daily IC difference)	Delta IC	NW 95% CI	t
ridge (ridge_a10000) - rule (rule_mom126vs)	-0.021	[-0.148, +0.106]	-0.33
lgb (lgb_l7_m1000) - ridge (ridge_a10000)	-0.028	[-0.123, +0.067]	-0.57
ensemble (ensemble) - lgb (lgb_l7_m1000)	+0.026	[-0.031, +0.083]	0.89

No rung's IC interval excludes zero, so by the course yardstick **no signal is established**. The pre-registered rule deploys the highest-IC rung, which is the **non-ML momentum rule**: Ridge (ridge_a10000, -0.011) and LightGBM (lgb_l7_m1000, -0.039) are both lower, and every LightGBM configuration has a negative IC (two significantly so). Early stopping chose 1-82 trees across folds, a symptom of very low signal-to-noise. ML adds nothing on this universe.

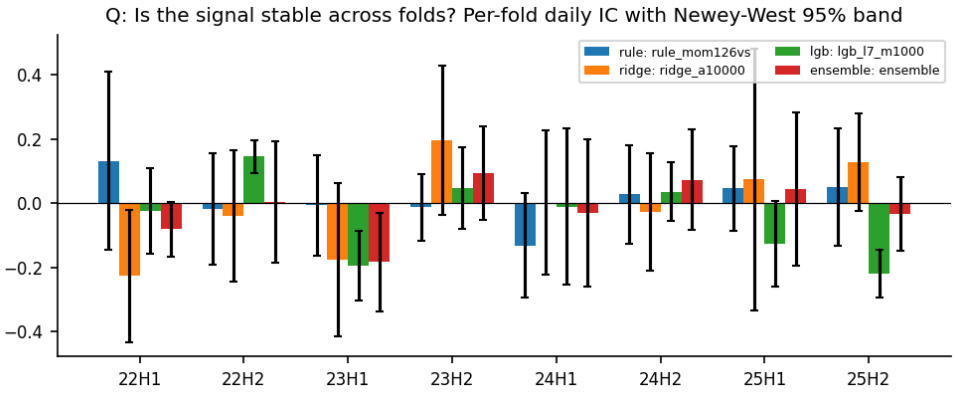


Figure M1. Q: Is the signal stable across folds? Per-fold IC with Newey-West bands; no rung is positive in more than 5 of 8 folds and every band spans zero in most folds.

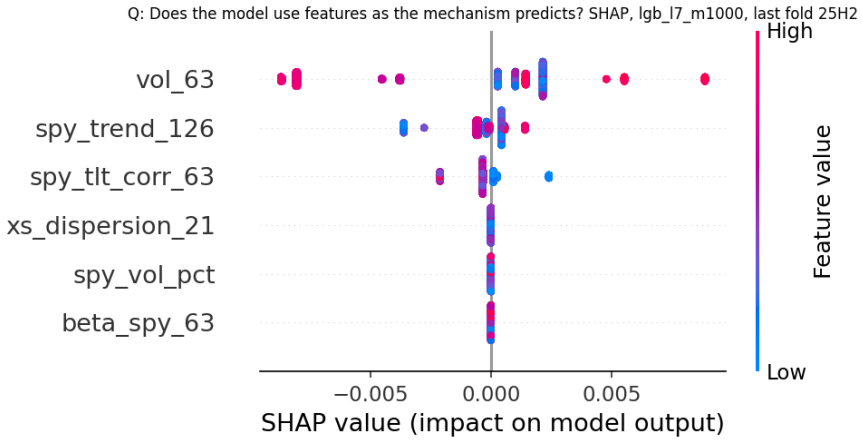


Figure 5 (required). Q: Does the ML candidate use features as the mechanism predicts? TreeSHAP for LightGBM (lgb_l7_m1000), last fold. The deployed rule uses one feature (vol-scaled momentum), so its attribution is trivial.

SHAP reading (four layers). Magnitude: beta_spy_63 and vol_63 carry the largest attribution; sign consistency: vol_63 has the same sign in 7/8 folds (high volatility lowers the score) while beta keeps its sign in only 3/8; stability: mean rank correlation of attributions across folds 0.44. **Surprising attribution:** the trees split on market-wide state variables that are identical for all ETFs on a date, so they cannot rank ETFs by themselves - LightGBM is fitting time variation, not the cross-section, which explains its negative IC. SHAP is descriptive, not causal (Ch. 15).

Fold	80% half-width (rank)	Coverage 80%	90% half-width	Coverage 90%	Flag < 80%
1	0.522	74.9%	0.696	91.2%	yes
2	0.565	80.2%	0.696	92.6%	no
3	0.565	81.0%	0.696	91.0%	no
4	0.565	78.0%	0.696	90.8%	yes
5	0.565	83.9%	0.696	90.9%	no
6	0.522	81.3%	0.696	93.1%	no
7	0.522	79.1%	0.652	90.9%	yes

Conformal intervals on the deployed score's cross-sectional rank (calibrated on the previous fold) are roughly calibrated (3 of 7 folds flagged below 80%) but uninformative: a half-width of about 0.57 on a -0.5..+0.5 scale covers most of the range, i.e. the score barely narrows an ETF's expected rank.

What could break this? With IC intervals spanning zero in validation and holdout, the evidence cannot distinguish prediction-quality drift from noise. A cross-section of 23 correlated ETFs gives few independent bets per date.

6. Backtest Protocol, Baseline, Portfolio, Costs, Risk

Component	Specification
Signal timing	Month-end close signal, next-open fill (no same-bar fills)
Rebalance and holding	Monthly; holding = 21-session label horizon
Position sizing	Equal weight top-5 (20% cap); inverse-vol comparison
Fill assumptions	Adjusted open; no fills on missing bars (assertion)
Constraints	Long-only, leverage <= 1, cash in BIL; asserted inside the loop
Costs	Per-share commission + tiered half-spread per instrument; 0-8x sweep; 3x spread stress

Validation (2022-2025) (net, excess of BIL)	Sharpe	CAGR	Vol	Max DD	Turnover 2-way/yr	Avg exposure
Deployed = rule + gate + vol target	0.36	6.9%	9.1%	-7.9%	9.0x	46.5%
Rule + gate, no overlay	0.28	7.2%	15.4%	-14.1%	11.4x	70.8%
Rule, no gate (+vol target)	0.41	8.4%	12.0%	-15.8%	7.7x	65.1%
Rule, no gate, no overlay	0.36	9.4%	20.2%	-24.7%	8.9x	100.0%
Ridge (ridge_a10000) + gate + vol target	0.16	4.9%	8.1%	-9.4%	5.6x	39.8%
LightGBM (lgb_l7_m1000) + gate + vol target	0.23	5.5%	7.9%	-7.5%	9.3x	39.8%
Ensemble + gate + vol target	0.14	4.8%	8.3%	-7.9%	7.0x	41.8%
Benchmark: equal-weight 23 ETFs	0.44	10.7%	17.9%	-20.6%	0.7x	100.0%
Benchmark: 60/40 SPY/AGG	0.29	6.8%	11.8%	-21.9%	0.4x	100.0%
Benchmark: SPY	0.46	11.2%	18.3%	-26.3%	0.3x	100.0%

Costs. Assumed cost 2.1 bp per leg; average holding 56 days. Break-even c* = gross excess return / two-way turnover: 31 bp per leg before the overlay and 36 bp with it (15x the assumed cost: comfortable zone). Costs are not what limits this strategy on validation; the signal is.

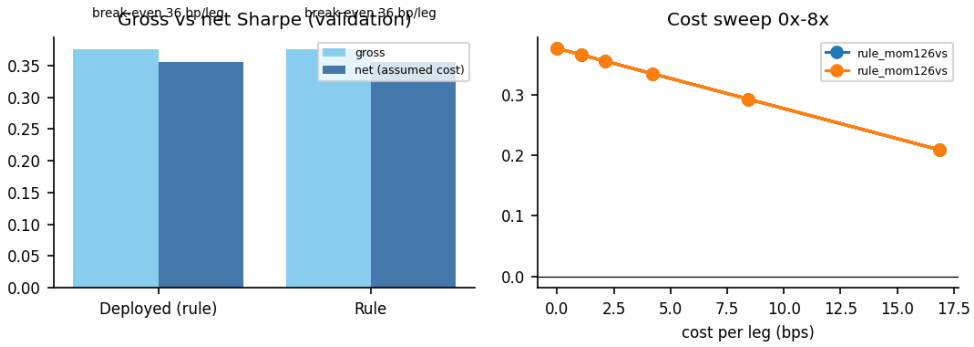


Figure B1. Q: How sensitive is the result to costs? Gross vs net and a 0-8x sweep (validation); the deployed strategy and the rule coincide.

Allocator (same signal, same 63-day window). Equal weight: Sharpe 0.28, MDD -14.1%, turnover 11.4x; inverse-vol: 0.27, -13.4%, 11.6x. Best-worst spread 0.003 Sharpe - far below the textbook's median 0.49; with five similar-volatility sector ETFs the allocator does not matter.

Risk control	Proxy	Control	Frequency
Market	Beta to SPY (rolling 126d; full-period HAC CI)	Vol-state gate + 10% vol target	Monthly decision, daily monitor
Factor / overlap	Count of tech-heavy names (QQQ, XLK, SOXX) held together	20% cap per name; disclosed overlap	Monthly
Leverage	Gross exposure	<= 1.0 asserted in the simulation	Every rebalance
Concentration	HHI / effective N	Max weight 20%	Monthly
Liquidity	Trailing median $ volume	>= $10M eligibility; ETF trade sizes << ADV	Monthly
Model	Rolling 63-day IC	Pause if 6-month IC < 0 (Section 8)	Monthly
Operational	Data checks; code hash vs freeze	Halt on unexplained nulls or hash mismatch	Daily

Tail and path risk. Daily VaR95 / CVaR95 -1.0% / -1.4%; by state: high-vol -1.0% / -1.4%, low-vol -0.9% / -1.3%. Max drawdown -7.9%, 57 sessions peak-to-trough, recovered after 96 sessions. Beta to SPY 0.08 [0.04, 0.11] (HAC): the strategy is mostly cash and a little equity.

Stress, spreads 3x: return / max DD	2022 rate shock	Aug-2024 carry unwind	Apr-2025 tariff shock
Deployed (= rule)	-0.7% / -5.1%	-2.6% / -4.6%	0.1% / -0.0%

Stress, spreads 3x: return / max DD	2022 rate shock	Aug-2024 carry unwind	Apr-2025 tariff shock
EW universe	-14.2% / -20.6%	-3.4% / -7.3%	-6.5% / -12.3%
SPY	-21.7% / -26.3%	-6.0% / -9.1%	-6.5% / -11.9%

Overlay and gate, with and without. Overlay: 0.28 -> 0.36 Sharpe, MDD -14.1% -> -7.9%, at an average exposure of only 46.5%. Gate on the rule (no overlay): 0.36 -> 0.28 - **the gate lowers Sharpe** while it cuts drawdown (-24.7% -> -14.1%); its value is defensive, as the stress table shows, not a better signal. **Kill-switch ladder:** -5% watch (PM), -10% halve (PM), -15% halt to BIL (risk owner), -20% unwind (team); on validation only the -5% level triggered (first 2022-09-27).

Statistical inference. Deployed validation Sharpe 0.36, analytic CI [-0.66, 1.37], bootstrap [-0.55, 1.30]; Deflated Sharpe 0.63 with N=60 (expected max Sharpe of zero-skill trials 0.18) and 0.58 with N=906; minimum track record to confirm this Sharpe: 49 years. Versus the equal-weight universe: -0.087 [-0.936, +0.725].

What could break this? The gate and stops act at month-end only, so an intra-month crash is not cut; the 3x-spread stress keeps linear costs; BIL as cash ignores its own small price risk.

7. Results, Diagnosis, and Next Iteration

Stage	Metric	Validation (2022-2025)	Holdout (2026-01..2026-10)	Benchmark (val / holdout)
Signal	Daily IC [NW 95% CI]; ICIR; share of positive folds (val) / months (holdout)	+0.010 [-0.057, +0.078]; 0.16; 4/8	-0.037 [-0.174, +0.099]; -; 44.4%	-
Portfolio (top-5 EW + gate, before costs and overlay)	Gross excess Sharpe [CI]; max DD; 2-way turnover/yr	0.29 [-0.55, 1.16]; -14.1%; 11.4x	0.13 [-1.94, 1.50]; -10.8%; 15.9x	EW universe 0.44 / 1.88; SPY 0.46 / 0.89; 60/40 0.29 / 0.61
Costs	Net excess Sharpe at assumed cost; break-even (bp per leg)	0.28; 31 (with overlay 36)	0.11; 5 (with overlay 4)	-
Risk overlay (10% vol target)	Net excess Sharpe; max DD	0.36; -7.9%	0.07; -6.4%	-
Inference	Deflated Sharpe (N=60 / N=906); minimum track record (years)	0.63 / 0.58; 49	0.46 / 0.43; 1242	-

Outcome pattern (Ch. 20): never resolved: the deployed Sharpe interval spans zero in validation and in the holdout. Confirmatory test: deployed minus equal-weight holdout Sharpe -1.816 [-3.602, +0.498], p = 0.15 -> **not passed**. Secondary: holdout IC -0.037 [-0.174, +0.099] -> not passed. Versus SPY -0.819 [-3.593, +0.688]. The gate was active in 2 of 10 holdout decisions. ML comparisons on the holdout (frozen, reported, not selected): LightGBM 0.02, ensemble 0.67, rule without gate 0.42.

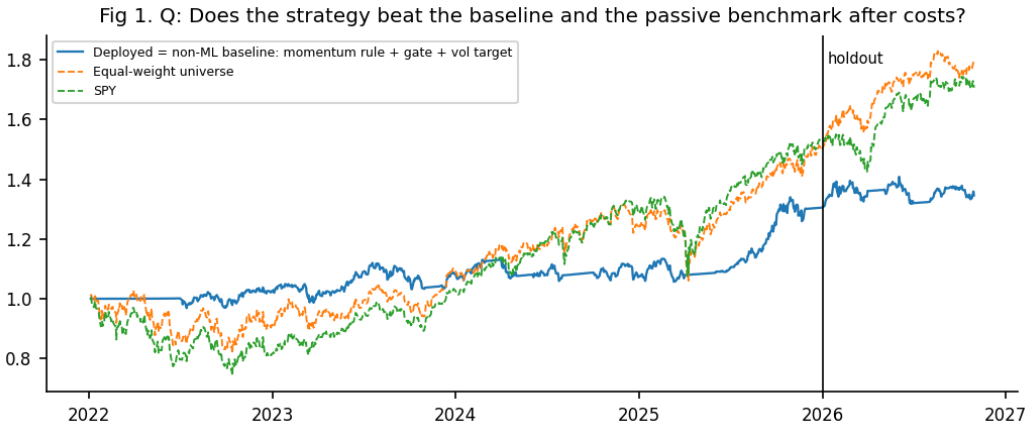


Figure 1. Q: Does the strategy beat the baseline and the passive benchmark after costs? Answer: no - the deployed strategy is the baseline, and it trails the equal-weight universe and SPY in validation and in the 2026 holdout; flat stretches are gated months in BIL.

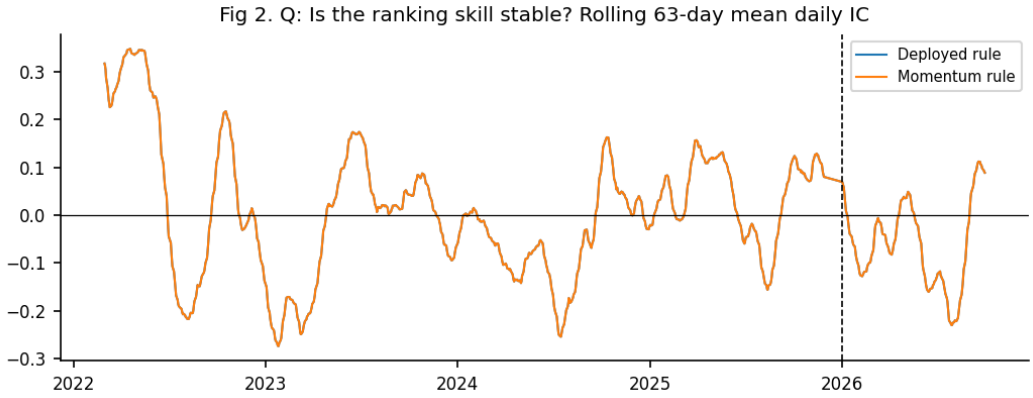


Figure 2. Q: Is the ranking skill stable? Answer: no - the rolling IC swings between about -0.27 and +0.35 and turns negative within weeks of the holdout start.

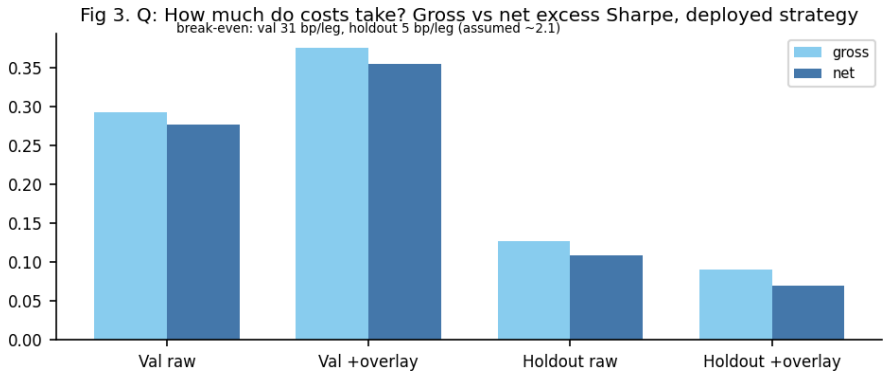


Figure 3. Q: How much do costs take? Answer: little - gross and net Sharpe are close; the weak signal, not costs, limits the strategy.

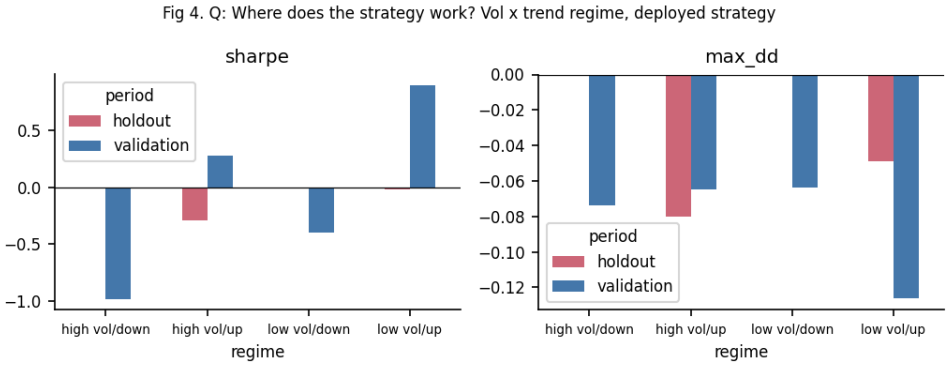


Figure 4. Q: Where does the strategy work? Answer: only in low-volatility up-trend months on validation (point-in-time thresholds); it loses in down-trends and is flat-to-negative in every holdout regime.

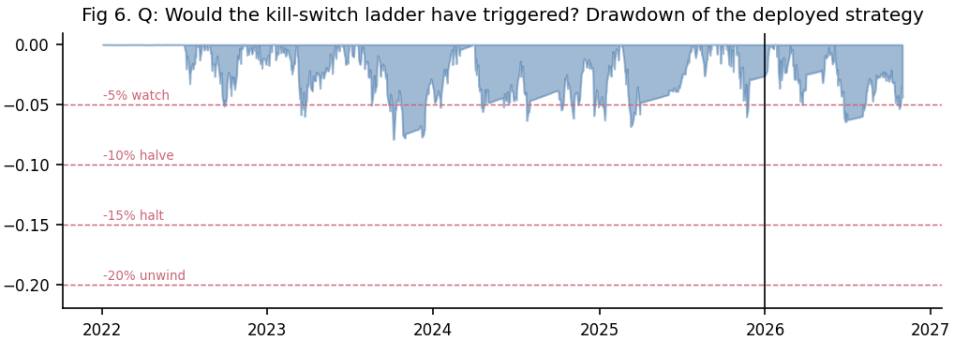


Figure 6. Q: Would the kill-switch ladder have triggered? Answer: only the -5% watch level, several times in validation and once in mid-2026; the overlay keeps drawdowns small because exposure averages under 50%.

Diagnosis. The strategy lost value at the **signal gate**: no rung established IC, momentum's IC is near zero overall and negative in calm states, and the holdout IC is -0.037. 2026 rewarded holding all sectors at once (equal-weight Sharpe 1.88, CAGR 24.8%); a top-5 rotation that is about half in cash cannot keep up in such a year. Drift diagnosis: prediction quality fell from +0.010 to -0.037 but both intervals span zero, and portfolio translation is stable (gross 0.09 vs net 0.07), so the gap is noise and regime, not a pipeline defect. Holdout break-even 5 bp per leg vs ~2.1 assumed: costs became material only because gross return collapsed. **Binding constraint:** label richness - 23 correlated names, monthly decisions and about 70 independent dates cannot resolve an IC of a few hundredths.

One next step. Keep the 23-ETF universe but lengthen the evidence before any new idea: extend the walk-forward backwards with the textbook's 2006-2019 history for these ETFs (about 25 more half-year folds) and re-test the momentum rule without the gate, since the gate was the part the data contradicted most clearly. A new holdout (2027 data) would be required. None of these first-pass strategies goes to capital.

What could break this? Ten holdout decisions; a single month moves the holdout Sharpe by several tenths. The holdout summary was computed by 06b from the saved single-touch artifacts after 06 failed at its final write step (Appendix A).

8. Monitoring and Deployment Plan (design, not built)

Parity. The same src/lib.py code path would run live in shadow mode for three months; compare stage by stage (raw bars, features, scores, target weights, orders, fills) against a backtest replay of the same days, and alert on any mismatch beyond rounding. **Monitoring.** Rolling 30/60/90-day Sharpe, drawdown, hit rate and daily IC; realisation ratio = live / backtest return over the same window (investigate below 0.5). **Drift.** PSI on the score input (vol-scaled 126-day momentum) and on the two risk features (beta, 63-day vol): watch 0.10-0.25, retrain above 0.25; monthly check of the top-5 holdings' overlap with the tech cluster (QQQ/XLK/SOXX). **Lifecycle.** Re-check every six months on the expanding window with the frozen protocol; pause if the 6-month IC is negative or the -15% kill-switch level is hit; retire if the pause lasts two consecutive reviews or the confirmatory test fails on a second fresh year. **Bugs vs decay vs costs (Ch. 26).** A parity mismatch or data-check failure is a bug (fix, rerun); falling IC with clean parity is statistical decay (retrain or retire); stable IC with falling net returns and rising fill slippage is cost inflation (slow the cadence or tighten the universe).

What could break this? Shadow mode cannot reveal capacity or market-impact problems; PSI thresholds are textbook defaults, not calibrated on this strategy.

Appendix A. Research Ledger, Run Log, Repository Map, Checklist

A.1 Research ledger (ledger/research_ledger.md, verbatim excerpt).

Research ledger

Taxonomy: Strategy -> Trial family -> Trial -> Run (ML4T Ch. 6). Every adaptive choice is a trial; bug fixes are not.

Prior work counted as trials (disclosed)

- v1 (textbook pipeline, 100 ETFs, 2006-2025): 734 registered backtests; its holdout 2024-2025 was run once.

- v2 (69 de-duplicated ETFs, 2020-2024): 52 strategy variants; its holdout 2024 was run once.

- v3 (54 de-duplicated ETFs, 2020-2025): 60 strategy variants; its holdout 2025 was run once.

All three are counted in the inclusive Deflated Sharpe trial count. Because 2024 and 2025 were already seen, they are development data here and the sealed holdout is 2026-01..2026-10, which no one on the team had looked at when this setup was committed.

The 23-ETF list was chosen by the team after those versions (hindsight risk in the universe choice, disclosed; it is not re-optimised here).

• 2026-11-03T18:51 (commit 7202343)

Setup v1 committed before any backtest. Mechanism: Industry and sector returns persist over one to six months because information diffuses slowly across industries and investors rebalance sector allocations gradually (industry momentum, flow predictability). The persistence crashes when market volatility spikes and leadership reverses abruptly, so the rotation signal should be traded only in calm-to-normal volatility states.

Source of edge: flow_predictability, risk_compensation. Holdout 2026-01-01..2026-10-30 sealed by the loader (src/lib.py: load_prices).

Pre-registered confirmatory test: holdout net Sharpe of deployed strategy minus equal-weight benchmark Sharpe (paired, 21-day block bootstrap); pass if 95% CI lower bound > 0. Secondary: holdout daily IC Newey-West 95% CI excludes zero.

Selection rule: highest mean validation IC; deployed strategy = selected rung + equal weight + gate + volatility target.

• 2026-11-03T18:51 (commit 7202343)

02 data: 27 symbols, 1760 sessions (dev). Duplicates 0, OHLC violations {'high_below_low': 0, 'close_outside_high_low': 0, 'open_outside_high_low': 0, 'non_positive_price': 0, 'zero_volume': 0}. PIT universe: 23-23 tradable names per decision. Survivorship check FAIL disclosed.

• 2026-11-03T18:51 (commit 7202343)

03 labels/features: label fwd_oo_21 (next-open to open+21). Nominal labelled rows 34,178 over 1486 dates; effective independent dates ~70. Searched set 15 signals + 4 states; merged by clustering: {'mom_126_skip21': 'mom_126_vs', 'sma50_ratio': 'ret_21', 'max_dd_63': 'vol_63', 'range_21': 'vol_63', 'rsi_14': 'ret_21'}. Triage: {'STOP': 13, 'REVISE': 2}. Model features (non-STOP signals + states): ['vol_63', 'beta_spy_63', 'spy_vol_pct', 'spy_trend_126', 'xs_dispersion_21', 'spy_tlt_corr_63']. Mechanism gate: momentum IC low-vol -0.013, high-vol +0.089.

• 2026-11-03T18:51 (commit 7202343)

04 models: ladder rule -> ridge (8 alphas, 1e-2..1e5) -> LightGBM (4 configs, L1 loss, early stopping on a purged inner split) -> ensemble. Chosen by validation IC: {'rule': 'rule_mom126vs', 'ridge': 'ridge_a10000', 'lgb': 'lgb_l7_m1000', 'ensemble': 'ensemble'}. Deployed rung (highest validation IC): rule. Paired deltas: ridge (ridge_a10000) - rule (rule_mom126vs): -0.0211 [-0.1484,+0.1061]; lgb (lgb_l7_m1000) - ridge (ridge_a10000): -0.0279 [-0.1231,+0.0674]; ensemble (ensemble) - lgb (lgb_l7_m1000): +0.0258 [-0.0312,+0.0829]

• 2026-11-03T18:52 (commit 7202343)

04 models: ladder rule -> ridge (8 alphas, 1e-2..1e5) -> LightGBM (4 configs, L1 loss, early stopping on a purged inner split) -> ensemble. Chosen by validation IC: {'rule': 'rule_mom126vs', 'ridge': 'ridge_a10000', 'lgb': 'lgb_l7_m1000', 'ensemble': 'ensemble'}. Deployed rung (highest validation IC): rule. Paired deltas: ridge (ridge_a10000) - rule (rule_mom126vs): -0.0211 [-0.1484,+0.1061]; lgb (lgb_l7_m1000) - ridge (ridge_a10000): -0.0279 [-0.1231,+0.0674]; ensemble (ensemble) - lgb (lgb_l7_m1000): +0.0258 [-0.0312,+0.0829]

• 2026-11-03T18:52 (commit 7202343)

Bug fix (not a trial): 04 conformal residuals used the raw deployed score; when the deployed rung is the rule its scale differs from the percentile target. Residuals now use the per-date percentile of the score. 04 re-run; model choices unchanged.

• 2026-11-03T18:52 (commit 7202343)

05 backtest (validation 2021-12-31..2025-12-31): 60 strategy variants. Deployed rule_mom126vs|ew|gate|vt: Sharpe 0.36, MDD -7.9%; rule rule_mom126vs|ew|gate|vt: 0.36; EW universe 0.44. Break-even 36 bp/leg. DSR(N=60) 0.63; DSR(N=906) 0.58.

FROZEN at 2026-11-03T18:52:39 (code sha256 362e5a0daf9f). Holdout not yet touched.

• 2026-11-03T18:53 (commit 9577d37)

06 traded the holdout once (2025-12-31..2026-10-30) and saved its artifacts, then failed while writing the summary JSON (deployed == baseline -> duplicate row). 06b computed the summary from the saved artifacts without re-running anything. Deployed rule_mom126vs|ew|gate|vt: excess Sharpe 0.07 [-2.01, 1.45]. Confirmatory test (minus EW): -1.82 [-3.60, +0.50] -> NOT PASSED. Holdout IC -0.037 [-0.174, +0.099].

A.2 Run log (run_log.csv): model:ridge: 128 rows, model:lgb: 64 rows, strategy: 60 rows, model:rule: 16 rows, feature_triage: 15 rows, holdout: 15 rows, setup: 1 rows, data: 1 rows, freeze: 1 rows. Each row stores timestamp, git commit, data snapshot hash (b8523e3640ed), setup version, trial family, trial, run, evaluation window,

configuration, metric and decision. Rows repeated by the 04 bug-fix re-execution are identified in the ledger.

A.3 Repository map

Path	Content
README.md	How to reproduce from the root; environment; why numbers could differ
setup.yaml	Trading setup v1 and protocol (Section 2)
data/build_dataset.py	Downloads data/prices_*.parquet from Yahoo (yfinance)
notebooks/01_setup ... 07_results (.py and .ipynb)	One stage each; 06 refuses to run twice; 06b summarises its saved output
src/lib.py	Sealed loader, PIT universe, features, statistics, simulator, run log
results/, figures/	Every number and figure in this document
run_log.csv, ledger/research_ledger.md	Run log and research ledger
requirements.txt, requirements.lock.txt	Environment (textbook uv env, Python 3.14)
report/, slides/	This PDF and the presentation, built by report/build_report.py and slides/build_slides.py

A.4 Pre-submission checklist (ten questions)

#	Question	Answer
1	Time each feature became knowable?	Yes: close of t for all features (prices/volume only)
2	Universe formed at t, leavers included?	Eligibility at t: yes. Fixed survivor list: disclosed FAIL
3	Label price convention matches fills?	Yes: next open to open+21
4	Scalers/winsorisation fitted on training folds only?	Yes (states); signals use per-date ranks
5	Buffers in trading days, every fold?	Yes: 21-session purge; embargo not needed (expanding, forward)
6	Holdout touched once, after the freeze?	Yes: freeze 2026-11-03T18:52:39, single touch recorded in results/holdout_touched.json (2026 data never seen before)
7	Ledger counts every adaptive choice; DSR uses it?	Yes: N=60 (this setup) and 906 (inclusive)
8	Per-instrument costs with sweep and break-even?	Yes: commission + tiered spread; 0-8x; break-even reported
9	Constraints and stops inside the simulation, t-1 info?	Yes: asserted in the loop; gate/overlay use the close of t, trade at t+1 open
10	Dataset and headline table regenerate from the root?	Yes: see README; LightGBM seed 42, bootstrap seed 7; Yahoo revisions disclosed